

SMEsPlus SaaS Foundation v0.4

Database Migration + API Schema + Screen Wireframe

1. Database Migration Baseline

1.1 Migration Order

1. tenants
 2. plans
 3. modules
 4. plan_modules
 5. companies
 6. branches
 7. divisions
 8. users
 9. roles
 10. permissions
 11. role_permissions
 12. user_role_assignments
 13. tenant_module_activations
 14. approval_workflows
 15. approval_requests
 16. notifications
 17. api_clients
 18. webhooks
 19. integration_events
 20. audit_logs
-

2. Core SQL Migration Draft

```
CREATE TABLE tenants (  
  id UUID PRIMARY KEY,  
  code VARCHAR(50) UNIQUE NOT NULL,  
  name VARCHAR(255) NOT NULL,  
  status VARCHAR(30) NOT NULL DEFAULT 'active',  
  plan_id UUID,  
  created_at TIMESTAMP NOT NULL DEFAULT now(),  
  updated_at TIMESTAMP NOT NULL DEFAULT now()  
);  
  
CREATE TABLE companies (  
  id UUID PRIMARY KEY,  
  tenant_id UUID NOT NULL REFERENCES tenants(id),  
  name VARCHAR(255) NOT NULL,  
  tax_id VARCHAR(50),
```

```
status VARCHAR(30) NOT NULL DEFAULT 'active',
created_at TIMESTAMP NOT NULL DEFAULT now(),
updated_at TIMESTAMP NOT NULL DEFAULT now()
);
```

```
CREATE TABLE branches (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  company_id UUID NOT NULL REFERENCES companies(id),
  code VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,
  address TEXT,
  status VARCHAR(30) NOT NULL DEFAULT 'active',
  created_at TIMESTAMP NOT NULL DEFAULT now(),
  updated_at TIMESTAMP NOT NULL DEFAULT now(),
  UNIQUE (tenant_id, company_id, code)
);
```

```
CREATE TABLE users (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  email VARCHAR(255) NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  status VARCHAR(30) NOT NULL DEFAULT 'active',
  mfa_enabled BOOLEAN NOT NULL DEFAULT false,
  created_at TIMESTAMP NOT NULL DEFAULT now(),
  updated_at TIMESTAMP NOT NULL DEFAULT now(),
  UNIQUE (tenant_id, email)
);
```

```
CREATE TABLE roles (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  name VARCHAR(100) NOT NULL,
  scope_type VARCHAR(30) NOT NULL,
  scope_id UUID,
  created_at TIMESTAMP NOT NULL DEFAULT now()
);
```

```
CREATE TABLE permissions (
  id UUID PRIMARY KEY,
  code VARCHAR(150) UNIQUE NOT NULL,
  module VARCHAR(100) NOT NULL,
  action VARCHAR(50) NOT NULL
);
```

```
CREATE TABLE role_permissions (
  role_id UUID NOT NULL REFERENCES roles(id),
  permission_id UUID NOT NULL REFERENCES permissions(id),
  PRIMARY KEY (role_id, permission_id)
);
```

```

);

CREATE TABLE user_role_assignments (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  user_id UUID NOT NULL REFERENCES users(id),
  role_id UUID NOT NULL REFERENCES roles(id),
  company_id UUID,
  branch_id UUID,
  created_at TIMESTAMP NOT NULL DEFAULT now()
);

CREATE TABLE modules (
  id UUID PRIMARY KEY,
  code VARCHAR(100) UNIQUE NOT NULL,
  name VARCHAR(255) NOT NULL,
  category VARCHAR(100),
  status VARCHAR(30) NOT NULL DEFAULT 'active'
);

CREATE TABLE tenant_module_activations (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  module_id UUID NOT NULL REFERENCES modules(id),
  status VARCHAR(30) NOT NULL DEFAULT 'active',
  activated_at TIMESTAMP NOT NULL DEFAULT now(),
  UNIQUE (tenant_id,module_id)
);

CREATE TABLE audit_logs (
  id UUID PRIMARY KEY,
  tenant_id UUID NOT NULL REFERENCES tenants(id),
  actor_id UUID,
  action VARCHAR(150) NOT NULL,
  resource_type VARCHAR(100) NOT NULL,
  resource_id UUID,
  ip_address VARCHAR(100),
  metadata JSONB,
  created_at TIMESTAMP NOT NULL DEFAULT now()
);

```

3. Row-Level Security Baseline

```

ALTER TABLE companies ENABLE ROW LEVEL SECURITY;
ALTER TABLE branches ENABLE ROW LEVEL SECURITY;
ALTER TABLE users ENABLE ROW LEVEL SECURITY;
ALTER TABLE roles ENABLE ROW LEVEL SECURITY;

```

```

ALTER TABLE user_role_assignments ENABLE ROW LEVEL SECURITY;
ALTER TABLE tenant_module_activations ENABLE ROW LEVEL SECURITY;
ALTER TABLE audit_logs ENABLE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation_companies
ON companies
USING (tenant_id = current_setting('app.tenant_id')::uuid);

CREATE POLICY tenant_isolation_branches
ON branches
USING (tenant_id = current_setting('app.tenant_id')::uuid);

CREATE POLICY tenant_isolation_users
ON users
USING (tenant_id = current_setting('app.tenant_id')::uuid);

CREATE POLICY tenant_isolation_audit_logs
ON audit_logs
USING (tenant_id = current_setting('app.tenant_id')::uuid);

```

4. API Schema Baseline

4.1 Standard Response

```

{
  "success": true,
  "data": {},
  "meta": {
    "request_id": "uuid",
    "tenant_id": "uuid",
    "timestamp": "2026-07-02T00:00:00Z"
  },
  "errors": []
}

```

4.2 Error Response

```

{
  "success": false,
  "data": null,
  "meta": {
    "request_id": "uuid",
    "timestamp": "2026-07-02T00:00:00Z"
  },
  "errors": [

```

```
{
  "code": "TENANT_ACCESS_DENIED",
  "message": "Tenant access denied",
  "field": null
}
]
```

4.3 Tenant Schema

```
Tenant:
  type: object
  properties:
    id:
      type: string
      format: uuid
    code:
      type: string
    name:
      type: string
    status:
      type: string
      enum: [active, suspended, inactive]
    plan_id:
      type: string
      format: uuid
    created_at:
      type: string
      format: date-time
```

4.4 Company Schema

```
Company:
  type: object
  properties:
    id:
      type: string
      format: uuid
    tenant_id:
      type: string
      format: uuid
    name:
      type: string
    tax_id:
      type: string
    status:
```

```
type: string
enum: [active, inactive]
```

4.5 User Schema

```
User:
type: object
properties:
  id:
    type: string
    format: uuid
  tenant_id:
    type: string
    format: uuid
  email:
    type: string
    format: email
  full_name:
    type: string
  status:
    type: string
    enum: [active, invited, suspended, inactive]
  mfa_enabled:
    type: boolean
```

4.6 Role Schema

```
Role:
type: object
properties:
  id:
    type: string
    format: uuid
  tenant_id:
    type: string
    format: uuid
  name:
    type: string
  scope_type:
    type: string
    enum: [tenant, company, branch, division]
  scope_id:
    type: string
    format: uuid
```

5. Wireframe หน้านำจ

5.1 Dashboard

```
+-----+
| SMEsPlus | Tenant Switcher | Company | User Menu |
+-----+
| Sidebar |
| - Dashboard |
| - Companies |
| - Users & Roles |
| - Modules |
| - Approvals |
| - Reports |
| - Audit |
+-----+
| KPI Cards |
| [Active Users] [Pending Approvals] [Active Modules] |
| |
| Approval Summary | Recent Activities |
+-----+ +-----+ |
| | Pending list | | Activity timeline | |
+-----+ +-----+ |
| |
| Module Shortcuts |
| [Accounting] [Sales] [Inventory] [HR] [CRM] |
+-----+
```

5.2 Tenant Administration

```
+-----+
| Tenant Administration |
+-----+
| [Create Tenant] [Search] [Filter: Status/Plan] |
+-----+
| Tenant Table |
| Code | Name | Plan | Status | Users | Actions |
+-----+
| T001 | ABC | Pro | Active | 45 | View/Edit |
+-----+
| Right Drawer: Tenant Detail |
| - Profile |
| - Plan |
| - Usage |
| - Health |
```

```

| - Suspend / Reactivate
+-----+

```

5.3 Company & Branch Management

```

+-----+
| Company & Branch Management
+-----+
| Left: Company List      Right: Company Detail
| +-----+ +-----+
| | Company A           | Profile
| | Company B           | Tax ID
| +-----+           | Status
|           +-----+
|           |
| Branches
| [Add Branch]
| Code | Name | Address | Status | Actions
+-----+

```

5.4 User & Role Management

```

+-----+
| User & Role Management
+-----+
| Tabs: [Users] [Roles] [Permissions] [Login History]
+-----+
| Users Tab
| [Invite User] [Search] [Filter Role/Status]
| Name | Email | Role | MFA | Status | Actions
+-----+
| User Detail Drawer
| - Profile
| - Assigned Roles
| - Scope: Tenant/Company/Branch
| - Deactivate User
+-----+

```


5.5 Module Activation

```
+-----+
| Module Activation                                |
+-----+
| Package: Professional                            |
| Billing Impact: ¥X / month                      |
+-----+
| Module Catalog                                  |
| [Accounting] Active                            |
| [Sales]      Active                            |
| [Inventory] Inactive [Activate]                |
| [HR]         Locked by Plan                    |
+-----+
| Activation Modal                                |
| - Dependency Check                             |
| - Billing Preview                               |
| - Confirm Activation                           |
+-----+
```

5.6 Configuration Center

```
+-----+
| Configuration Center                            |
+-----+
| Sidebar Settings                               |
| - General                                      |
| - Fiscal Year                                 |
| - Tax                                          |
| - Numbering Series                           |
| - Approval Rules                             |
| - Notifications                              |
+-----+
| Main Panel                                     |
| Setting Form                                  |
| [Save Changes] [View Change History]          |
+-----+
```

5.7 Approval Inbox

```
+-----+
| Approval Inbox                                  |
+-----+
```

Filters: Status Module Requester Date	
+-----+	
Pending Approval List	
Request No Module Requester Amount Status	
+-----+	
Approval Detail	
- Request Summary	
- Attachment	
- Approval Timeline	
- Comment	
[Approve] [Reject] [Delegate]	
+-----+	

5.8 Integration Center

+-----+	
Integration Center	
+-----+	
Tabs: [API Clients] [Webhooks] [Event Logs]	
+-----+	
API Clients	
Name Status Created Last Used Actions	
[Create API Client]	
+-----+	
Webhooks	
URL Event Status Retry Actions	
+-----+	

5.9 Audit & Governance

+-----+	
Audit & Governance	
+-----+	
Search / Filter	
User Action Resource Date IP	
+-----+	
Audit Table	
Time Actor Action Resource IP Detail	
+-----+	
Detail Drawer	
- Metadata JSON	
- Before/After	
- Request ID	

| - Export

5.10 Reports

```
+-----+
| Reports                                     |
+-----+
| Report Catalog                             |
| [Usage Report] [User Activity] [Audit Report] |
+-----+
| Filter Panel                               |
| Date Range | Company | Branch | Module    |
+-----+
| Preview Table / Chart                     |
|                                           |
| [Export PDF] [Export Excel] [Schedule Report] |
+-----+
```

6. Next Development Output

หลัง v0.4 สามารถแตกเอกสารต่อเป็น:

1. 001_initial_schema.sql
2. 002_rls_policies.sql
3. openapi.foundation.yaml
4. screen-spec-foundation.md
5. foundation-mvp-backlog.md

Status

SMEsPlus Foundation อยู่ในสถานะ:

READY FOR:

- Backend scaffold
- Database migration draft
- OpenAPI mock
- Frontend wireframe
- Sprint backlog refinement

ยังควรทำต่อก่อน production:

- Full OpenAPI request/response ราย endpoint
- Complete ERD diagram

- UI component design system
- Security test cases
- Tenant isolation automated tests